Problem - 01

* $3x + 5y - 7z = 13$

$4x + y - 12z = 6$

$2x + 9y - 3z = 20$

Solution:

The given system of linear equation is,

$$\left.\begin{aligned} 3x + 5y - 7z &= 13 \\ 4x + y - 12z &= 6 \\ 2x + 9y - 3z &= 20 \end{aligned}\right\} \quad -(1)$$

The system (1) can be written as.

$$AX = B$$

where, $A = \begin{pmatrix} 3 & 5 & -7 \\ 4 & 1 & -12 \\ 2 & 9 & -3 \end{pmatrix}$, $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ And $B = \begin{pmatrix} 13 \\ 6 \\ 20 \end{pmatrix}$

The augmented matrix of the system is,

$$[A|B] = \left(\begin{array}{ccc|c} 3 & 5 & -7 & 13 \\ 4 & 1 & -12 & 6 \\ 2 & 9 & -3 & 20 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 3 & 5 & -7 & 13 \\ 0 & 17 & 8 & 34 \\ 2 & 9 & -3 & 20 \end{array}\right) \quad R_2' = 4R_1 - 3R_2$$

$$\sim \left(\begin{array}{ccc|c} 3 & 5 & -7 & 13 \\ 0 & 17 & 8 & 34 \\ 0 & 17 & 5 & 34 \end{array}\right) \quad R_3' = 3R_3 - 2R_1$$

$$\sim \left(\begin{array}{ccc|c} 3 & 5 & -7 & 13 \\ 0 & 17 & 8 & 34 \\ 0 & 0 & 3 & 0 \end{array}\right) \quad R_3' = R_2 - R_1$$

which is the echelon form of the augmented matrix.

Now,

$$P(A|B) = P(A)$$

so, the solution of the linear equation exists.

& No. of variable = Rank

$$3 = 3$$

So, the system is consistent and has unique solution

The reduced system is,

$$3x + 5y - 7z = 13$$

$$17y - 8z = 34$$

$$3z = 0$$

By Back substitution we get,

$$z = 0$$

$$17y - 8z = 34 \Rightarrow 17y - 8.0 = 34 \Rightarrow y = 2$$

$$3x + 5y - 7z = 13 \Rightarrow 3x + 5.2 - 7.0 = 13 \Rightarrow x = 1$$

Hence, the required result is,

$$x = 1 \; ; \; y = 2 \; ; \; z = 0$$

Problem = 2

$$2x - 4y + 6z = 8$$
$$3x - 2y + z = 4$$
$$-x + 2y - 3z = -4$$

Solution:

The given system of linear equation is.

$$\left.\begin{aligned} 2x - 4y + 6z &= 8 \\ 3x + 2y + z &= 4 \\ -x + 2y - 3z &= -4 \end{aligned}\right\} \text{---} (1)$$

The system can be written as,

$$AX = B$$

where, $A = \begin{pmatrix} 2 & -4 & 6 \\ 3 & 2 & 1 \\ -1 & 2 & -3 \end{pmatrix}$, $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ And $B = \begin{pmatrix} 8 \\ 4 \\ -4 \end{pmatrix}$

The augmented matrix of the given system is.

$$[A|B] = \left(\begin{array}{ccc|c} 2 & -4 & 6 & 8 \\ 3 & 2 & 1 & 4 \\ -1 & 2 & -3 & -4 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 2 & -4 & 6 & 8 \\ 0 & 16 & -16 & -16 \\ -1 & 2 & -3 & -4 \end{array}\right) \quad R_2' = 2R_2 - 3R_1$$

$$\sim \left(\begin{array}{ccc|c} 2 & -4 & 6 & 8 \\ 0 & 16 & -16 & -16 \\ 0 & 0 & 0 & 0 \end{array}\right) \quad R_3' = 2R_3 + R_1$$

which is the echelon form of the augmented matrix.

Now,
$$P(A|B) = P(A)$$

So the solution of the linear equation exists.

And No. of variable $\neq$ Rank

$\Rightarrow$, $3 \neq 2$

So, the system has many solutions

The reduced system is,

$$2x - 4y + 6z = 8$$

$$16y - 16z = -16$$

$$0x + 0y + 0z = 0$$

Since, $0=0$ occurs let's put

$$z = t$$

$0.x + 16y - 16t = -16 \Rightarrow y = t - 1$

$2x - 4(t-1) + 6.t = 8 \Rightarrow x = \frac{1}{2}[8 + 4(t-1) - 6t] \Rightarrow x = 2 - t$

General solutions are

$$x = 2 - t \,;\, y = t - 1 \,,\, z = t$$

So the required result is,

$$x = 2 - t \,;\; y = t - 1 \,;\; z = t$$

* $2x + 3y + 5z + t = 3$

$3x + 4y + 2z + 3t = -2$

$x + 2y + 8z - t = 8$

$7x + 9y + z + 8t = 0$

<u>Solution:</u>

The given system of linear equation is,

$$\left.\begin{aligned} 2x + 3y + 5z + t &= 3 \\ 3x + 4y + 2z + 3t &= -2 \\ x + 2y + 8z - t &= 8 \\ 7x + 9y + z + 8t &= 0 \end{aligned}\right\} \text{--- } (1)$$

the system (1) can be written as,

$$AX = B$$

where, $A = \begin{pmatrix} 2 & 3 & 5 & 1 \\ 3 & 4 & 2 & 3 \\ 1 & 2 & 8 & -1 \\ 7 & 9 & 1 & 8 \end{pmatrix}$, $X = \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix}$ And $B = \begin{pmatrix} 3 \\ -2 \\ 8 \\ 0 \end{pmatrix}$

$$[A|B] = \left(\begin{array}{cccc|c} 2 & 3 & 5 & 1 & 3 \\ 3 & 4 & 2 & 3 & -2 \\ 1 & 2 & 8 & -1 & 8 \\ 7 & 9 & 1 & 8 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 8 & -1 & 8 \\ 3 & 4 & 2 & 3 & -2 \\ 2 & 3 & 5 & 1 & 3 \\ 7 & 9 & 1 & 8 & 0 \end{array}\right) \quad R_1 \leftrightarrow R_3$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 8 & -1 & 8 \\ 0 & -2 & -22 & 6 & -26 \\ 2 & 3 & 5 & 1 & 3 \\ 7 & 9 & 1 & 8 & 0 \end{array}\right) \quad R_2' = R_2 - 3R_1$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 8 & -1 & 8 \\ 0 & -2 & -22 & 6 & -26 \\ 0 & -1 & -11 & 3 & -13 \\ 7 & 9 & 1 & 8 & 0 \end{array}\right) \quad R_3' = R_3 - 2R_1$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 8 & -1 & 8 \\ 0 & -2 & -22 & 6 & -26 \\ 0 & -1 & -11 & 3 & -13 \\ 0 & -5 & -55 & 15 & -56 \end{array}\right) \quad R_4' = R_4 - 7R_1$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 8 & -1 & 8 \\ 0 & -2 & -22 & 6 & -26 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & -5 & -55 & 15 & -56 \end{array}\right) \quad R_3' = 2R_3 - R_2$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 8 & -1 & 8 \\ 0 & -2 & -22 & 6 & -26 \\ 0 & 0 & 0 & 0 & 18 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \quad \begin{array}{l} R_4' = 2R_4 - 5R_2 \\ R_4 \Leftrightarrow R_3 \end{array}$$

which is the echelon form of the augmented matrix.

Now, $P(A|B) = 4 \neq P(A) = 3$

And, No. of variable $\neq$ Rank

$4 \neq 3$

So, the system is inconsistent and has no solution.

The reduced system is,

$$\left.\begin{aligned} x + 2y + 8z - t &= 8 \\ y + 11z - 3t &= 13 \\ 0 &= 0 \\ 0 &= 18 \end{aligned}\right\}$$

Since, $0 = 18$ occurs, which is not possible. So the given system of linear equation is inconsistent and has no solution